7.2.2 We find $n_x = 5$, $n_y = 8$, $\bar{x} = 539.2$, $\bar{y} = 544.625$, $s_x^2 = 4948.7$, $s_y^2 = 4327.98$, $s_p = \sqrt{\frac{(n_x-1)s_x^2 + (n_y-1)s_y^2}{n_x+n_y-2}} = 67.481$,

$$\bar{x} - \bar{y} \pm t_{\alpha/2}(n_x+n_y-2)\, s_p \sqrt{\frac{1}{n_x} + \frac{1}{n_y}} = -5.425 \pm 1.796(67.481)\sqrt{\frac{1}{5}+\frac{1}{8}}$$

$(\alpha/2 = 0.05)$

$= \boxed{-5.425 \pm 69.09 = [-74.52, 63.67]}$.

7.3.10 (a) $\hat{p}_1 = y_1/n_1 = 28/194 = 0.1443$

(b) Using equation (7.3-2), $\hat{p}_1 \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}} = 0.1443 \pm 1.96\,(0.02523)$

$= \boxed{0.1443 \pm 0.04945 = [0.09485, 0.1938]}$

(c) $\hat{p}_1 - \hat{p}_2 = y_1/n_1 - y_2/n_2 = 0.07643$.

(d) As on page 327, a lower bound is $\hat{p}_1 - \hat{p}_2 - z_\alpha\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

$= 0.07643 - 1.645\,(0.032) = 0.0238$. So the interval is

$\boxed{[0.0238, 1]}$.

7.3.12 (a) As in (d) in the previous question (but now for the 2-sided case), our interval is $\hat{p}_A - \hat{p}_B \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_A(1-\hat{p}_A)}{n_1} + \frac{\hat{p}_B(1-\hat{p}_B)}{n_2}}$

$= \boxed{0.0491 \pm 0.06203 = [-0.0129, 0.111]}$

7.4.7 Since we don't have prior info. about p, we use equation (7.4-4): $n = \dfrac{z_{\alpha/2}^2}{4\varepsilon^2}$.

ⓐ $\varepsilon = 0.03$, $z_{\alpha/2} = 1.96 \Rightarrow n = 1067.1 \nearrow \boxed{1068}$

ⓑ $\varepsilon = 0.02$, $z_{\alpha/2} = 1.96 \Rightarrow n = \boxed{2401}$

ⓒ $\varepsilon = 0.03$, $z_{\alpha/2} = 1.645 \Rightarrow n = 751.7 \nearrow \boxed{752}$

Problem 5 $W = \bar{X} - \bar{Y}$ is $N(\mu_x - \mu_Y, \frac{3^2}{n} + \frac{4^2}{n}) = N(\mu_x - \mu_Y, \frac{5^2}{n})$, hence $P(\mu_x - \mu_Y \leq \bar{X} - \bar{Y} + 0.5) = P(-0.5 \leq \bar{X} - \bar{Y} - (\mu_x - \mu_Y))$
$= P\left(\sqrt{n}\left(\frac{-0.5}{5}\right) \leq Z\right)$, where $Z \sim N(0,1)$. Now $P(\sqrt{n}(-0.1) \leq Z) = P(Z \leq (0.1)\sqrt{n})$, and this should be ≥ 0.9, i.e. $P(Z > (0.1)\sqrt{n}) \leq 0.1$ (since $P(Z \leq 0.1(\sqrt{n})) = 1 - P(Z > 0.1(\sqrt{n}))$).
Thus $(0.1)\sqrt{n} \geq z_{0.1} = 1.282$, or $n \geq (12.82)^2 = 164.4$, or $\boxed{n \geq 165}$.

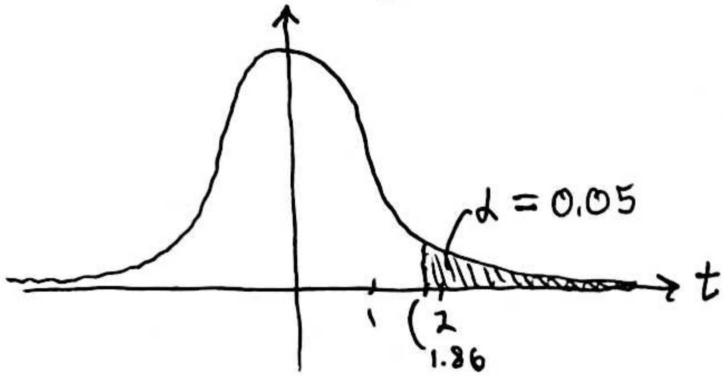
This was from scratch; you could also just use the formula $n = \dfrac{z_\alpha^2 \sigma^2}{\varepsilon^2}$ (modified equation (7.4-1)), where $\sigma = \sqrt{3^2+4^2} = 5$, $\varepsilon = 0.50$, $z_\alpha = 1.282$.

8.1.6 ⓐ $H_0: \mu = 3.4$ (μ = mean FVC of volleyball players)

ⓑ $H_1: \mu > 3.4$

ⓒ $T = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}}$, $\mu_0 = 3.4$

ⓓ Given H_0, T has a t-distribution with $n-1$ degrees of freedom. Using the table 8.1-2 on page 365, a critical region is $t \ge t_\alpha(n-1) = t_{0.05}(8) = 1.86$, or $\bar{x} \ge 3.4 + 1.86 \cdot s/\sqrt{9}$. In terms of t, this looks like



(e) $\bar{x} = 3.556$, $s = 0.1667 \Rightarrow t = \dfrac{\bar{x} - 3.4}{s/\sqrt{9}} = 2.807$.

(f) Since $2.807 > 1.86$, we can be 95% confident in rejecting H_0, i.e. in concluding $\mu > \mu_0 = 3.4$.

(g) p-value $= P(T \ge 2.807;\ \mu = 3.4)$. From the table, $t_{0.025}(8) = 2.306$ and $t_{0.01}(8) = 2.896$, so the p-value is between 0.01 and 0.025.

<u>8.1.12</u> $\bar{x}_D = \dfrac{81}{17} = 4.765$, $s_D = 9.087$, $t_{0.05}(16) = 1.746$,

$t = \dfrac{\bar{x}_D}{s_D/\sqrt{17}} = 2.162 > 1.746 \Rightarrow$ at 95% confidence level, A goes farther than B.